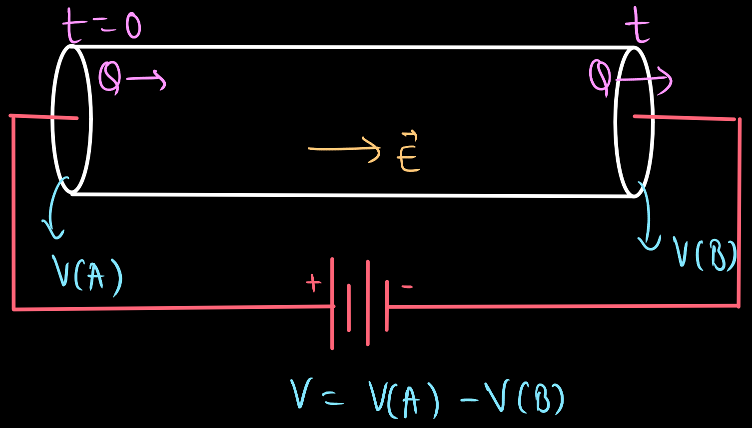
Electric Energy and Power

* Consider a conductor with end points A and B, in which a current I is flowing from A to B.
* The electric potential at A and B are denoted by V (A) and V (B) respectively.
* Since current is flowing from A to B, V (A) > V (B) and the potential difference across AB is

$$V = V(A) - V(B) > 0.$$



* In a time interval t, an amount of charge $Q = I\,t$ travels from A to B. The potential energy of the charge at A, by definition, was $Q\,V(A)$ and similarly at B, it is $Q\,V(B)$.
* Thus, change in its potential energy ΔU is

ΔU = Final potential energy − Initial potential energy = $Q\,V(B) - QV(A)$

$\Delta U = -Q\,[\,V(A) - V(B)] = -\,Q\,V = -\,ItV$

* In case charges were moving freely through the conductor under the action of electric field, their kinetic energy would increase as they move. $\Delta K = -\,\Delta U = VIt$.
* However, seen earlier that on the average, charge carriers do not move with acceleration but with a steady drift velocity. This is because of the collisions with ions and atoms during transit.
* During collisions, the energy gained by the charges thus is shared with the atoms. The atoms vibrate more vigorously, i.e., the conductor heats up.

$$\boxed{\begin{aligned} H &= -\,\Delta U = VIt \\ H &= I^2\,R\,t = V^2t/R \end{aligned}}$$

amount of energy dissipated as heat in the conductor during the time interval t

* Power : The energy dissipated per unit time is the power dissipated

$$P = H/t$$

$$\boxed{P = VI = I^2R = V^2/R}$$

S.I Unit of Power is Watt (W)

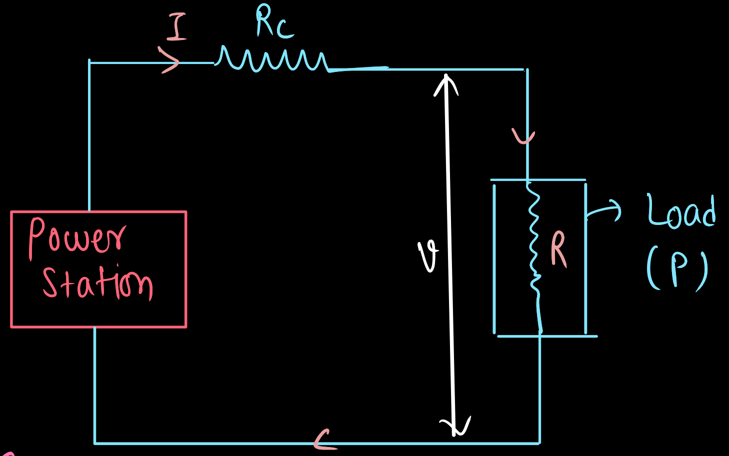
* Power loss in Transmission Cabels :

Consider a device R, to which a power P is to be delivered via transmission cables having a resistance Rc

If V is the voltage across R and I the current through it, then

$$P = VI \rightarrow I = P/V$$

The connecting wires from the power station to the device has a finite resistance Rc. The power (Pc) dissipated in the connecting wires.



$$P_c = I^2 R_c$$

$$\boxed{P_c = P^2 R_c / V^2}$$

* Thus, to drive a device of power P, the power wasted in the connecting wires is inversely proportional to V^2. To reduce Pc , these wires carry current at high voltages.
* Using electricity at such voltages is not safe and hence at the other end, a device called a transformer lowers the voltage to a value suitable for use.

21. (a) An electric iron rated 2·2 kW, 220 V is operated at 110 V supply. Find : *2*

(i) its resistance, and

(ii) heat produced by it in 10 minutes. CBSE 26

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(i) $R=\dfrac{V^2}{P}$	
$=\dfrac{(220)^2}{2.2 \times 10^3}=22\Omega$	½
(ii) $H = I^2 R t$	½
$I=\dfrac{V_{applied}}{R}=\dfrac{110}{22}=5A$	½
$=5 \times 5 \times 22 \times 600$	
$=3.3 \times 10^5$ J	½

16. **Assertion (A)** : Two electric heaters of power P_1 and P_2 ($> P_1$) are joined in series across a dc source of voltage V. The power consumed by the combination will be less than that consumed by P_1 when connected across the same source. **1**

Reason (R) : The power consumed by a electric device when connected to a dc source of voltage V is proportional to its resistance.

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(C) Assertion (A) is true but Reason (R) is false

11. Two heaters rated as (P_1, V) and (P_2, V) are connected in series across a dc source of $\frac{V}{2}$ volt. The power consumed by the combination will be – **1**

(A) $(P_1 + P_2)$

(B) $\frac{P_1 + P_2}{2}$

(C) $\frac{P_1 P_2}{2(P_1 + P_2)}$

(D) $\frac{P_1 P_2}{4(P_1 + P_2)}$

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(D) $\frac{P_1 P_2}{4(P_1 + P_2)}$

17. *Assertion (A) :* When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.

Reason (R) : In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

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Reason (R) : In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

9. A conductor of 10 Ω is connected across a 6 V ideal source. The power supplied by the source to the conductor is **1**

(a) 1.8 W (b) 2.4 W

(c) 3.6 W (d) 7.2 W

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3. A current of 0.8 A flows in a conductor of 40 Ω for 1 minute. The heat produced in the conductor will be **1**

(a) 1445 J (b) 1536 J

(c) 1569 J (d) 1640 J

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25. (a) When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady ? Why ? *2*

OR

(b) A battery is connected first across the series combination and then across the parallel combination, of three resistances R, 2R and 3R. In which of the three resistances will power dissipated be maximum in the two cases ? Justify your answer. *2*

(a) For Answer	1
Reason	1

For series Answer	½
Justification	½
For parallel Answer	½
Justification	½

(a) Power delivered just after it is turned on, will be more because resistance of the bulb is low. After some time temperature of the bulb increases and resistance also increases and therefore power $(\frac{V^2}{R})$ becomes low.	1 1
For series Power dissipated will be maximum for 3R. Because current is same and power is proportional to resistance. For parallel Power dissipated will be maximum for R. Because voltage is same and power is inversely proportional to resistance.	 ½ ½ ½ ½

1. A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be

(A) P

(B) $\frac{P}{2}$

(C) $\frac{P}{4}$

(D) 2P

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20. Two electric heaters have power ratings P_1 and P_2, at voltage V. They are connected in series to a dc source of voltage V. Find the power consumed by the combination. Will they consume the same power if connected in parallel across the same source ? **2**

Finding power consumed by two electric heaters in series combination	1 ½
Writing answer for parallel combination	½

$R_1 = \dfrac{V^2}{P_1}$ & $R_2 = \dfrac{V^2}{P_2}$	½
$R_{eq} = R_1 + R_2 = V^2\left(\dfrac{1}{P_1} + \dfrac{1}{P_2}\right)$	½
$P_{series} = \dfrac{V^2}{R_{eq}}$	
$P_{series} = \dfrac{V^2}{V^2\left(\dfrac{1}{P_1} + \dfrac{1}{P_2}\right)}$	
$\dfrac{1}{P_{series}} = \dfrac{1}{P_1} + \dfrac{1}{P_2}$	½
No	½

3. A steady current I is passed through a conductor at room temperature for time t. It is observed that its temperature rises by 0·5°C. If 2I current is passed through the conductor (at room temperature) for the same duration, the rise in its temperature will be approximately :

(A) 1·0°C (B) 1·5°C

(C) 2·0°C (D) 4·0°C

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(A) $1{\cdot}0°C$ (B) $1{\cdot}5°C$

(C) $2{\cdot}0°C$ (D) $4{\cdot}0°C$

(C) $2.0\ ^0C$

1. A wire of resistance R, connected to an ideal battery consumes a power P. If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be :

(A) $\frac{P}{4}$

(B) $\frac{P}{2}$

(C) P

(D) 2P

1. A wire of resistance R, connected to an ideal battery consumes a power P. If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be :

(A) $\frac{P}{4}$

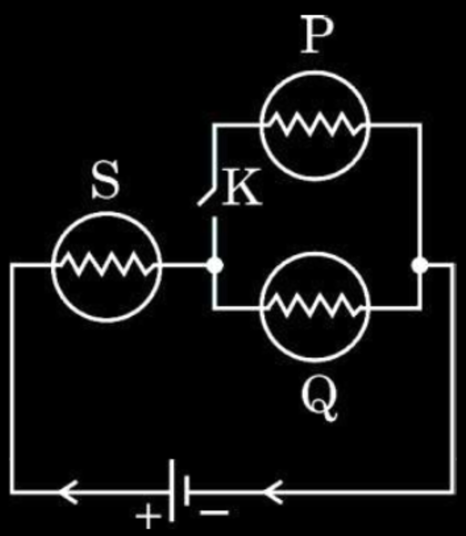
(B) $\frac{P}{2}$

(C) P

(D) 2P

(A) $\frac{P}{4}$

21. (a) In the given figure, three identical bulbs P, Q and S are connected to a battery.



(i) Compare the brightness of bulbs P and Q with that of bulb S when key K is closed.

(ii) Compare the brightness of the bulbs S and Q when the key K is opened.

Justify your answer in both cases. *2*

(i) Comparison of brightness of bulbs P and Q with bulb S	½
Justification	½
(ii) Comparison of brightness of bulb S with Q	½
Justification	½

(i) Brightness of the bulb ‘S’ will be more than bulbs ‘P’ and ‘Q’	½
The current flowing through the bulb ‘S’ is twice of the current in bulbs ‘P’ and ‘Q’.	½
(ii) Brightness of the bulb ‘S’ and ‘Q’will be same	½
The current flowing through both bulbs is same.	½
Alternatively-	
(i) Brightness of the bulb ‘S’ will be more than bulbs ‘P’ and ‘Q’	
The potential difference across ‘S’ is twice than the potential difference across bulbs ‘P’ and ‘Q’	
(ii) Brightness of both bulbs ‘S’ and ‘Q’ is same.	
The potential difference across ‘S’ and ‘Q’ will be same.	

6. The element of a heater is rated (P, V). If it is connected across a source of voltage $\frac{V}{2}$, then the power consumed by it will be *1*

(A) P

(B) 2P

(C) $\frac{P}{2}$

(D) $\frac{P}{4}$

6. The element of a heater is rated (P, V). If it is connected across a source of voltage $\frac{V}{2}$, then the power consumed by it will be *1*

(A) P

(B) 2P

(C) $\frac{P}{2}$

(D) $\boxed{\frac{P}{4}}$

4. Two resistors R_1 and R_2 of $4\ \Omega$ and $6\ \Omega$ are connected in parallel across a battery. The ratio of power dissipated in them, $P_1 : P_2$ will be *1*

(A) $4 : 9$

(B) $3 : 2$

(C) $9 : 4$

(D) $2 : 3$

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(A) 4 : 9

$\boxed{\text{(B)} \quad 3 : 2}$

(C) 9 : 4

(D) 2 : 3

Two bulbs are rated (P_1, V) and (P_2, V). If they are connected (i) in series and (ii) in parallel across a supply V, find the power dissipated in the two combinations in terms of P_1 and P_2.

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$R_1 = \frac{V^2}{P_1}$, $R_2 = \frac{V^2}{P_2}$,	½
$P_s = \frac{V^2}{Rs} = \frac{P_1 P_2}{P_1+P_2}$	
$\frac{1}{Ps} = \frac{1}{P_1} + \frac{1}{P_2}$	½
$\frac{1}{Rp} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{P_1+P_2}{V^2}$	½
$\therefore\ P_p = \frac{V^2}{R_P} = P_1 + P_2$	½

Two electric bulbs P and Q have their resistances in the ratio of 1 : 2. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs.

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Power = $I^2 R$ The current, in the two bulbs, is the same as they are connected in series.	½ ½
$\therefore \dfrac{P_1}{P_2} = \dfrac{I^2 R_1}{I^2 R_2} = \dfrac{R_1}{R_2}$	½
$= \dfrac{1}{2}$	½

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Nichrome and copper wires of same length and same radius are connected in series. Current I is passed through them. Which wire gets heated up more ? Justify your answer.

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i.	Nichrome	½
ii.	$R_{Ni} > R_{Cu}$ (or $Resistivity_{Ni} > Resistivity_{Cu}$)	½

(a) The potential difference applied across a given resistor is altered so that the heat produced per second increases by a factor of 9. By what factor does the applied potential difference change ?

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a) $H = \frac{V^2}{R}$	½
$\therefore$ V increases by a factor of $\sqrt{9} = 3$	½